ISTANBUL TECHNICAL UNIVERSITY
FACULTY OF COMPUTER AND
INFORMATICS

TinyNPU

Graduation Project Final Report

Fırat Kızılboğa
150200115

Department: Computer Engineering
Division: Computer Engineering

Advisor: Assoc. Prof. Dr. Ayse Yilmazer

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Statement of Authenticity

I/we hereby declare that in this study

1. all the content influenced from external references are cited clearly and in detail,
2. and all the remaining sections, especially the theoretical studies and implemented software/hardware that constitute the fundamental essence of this study is originated by my/our individual authenticity.

İstanbul, Mart 2026

Fırat Kızılboğa

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TinyNPU

(SUMMARY)

TinyNPU is a compact accelerator implemented as a complete hardware/software stack. It combines a parameterized systolic-array neural processing unit with a PyTorch-first segmented compiler and runtime, a quantization-aware model-preparation flow, RTL and host-side verification infrastructure, and performance and synthesis characterization tooling. The stack is designed to make the datapath usable from a realistic machine-learning workflow and measurable in a disciplined way.

The hardware architecture is organized around an $N \times N$ output-stationary systolic array, a unified buffer, an instruction memory, streamer logic that feeds the array, and a post-processing unit that applies bias, rescaling, activation, and packed writeback. The software stack traces PyTorch models, partitions them into accelerator segments and host operations, compiles supported segments into TinyNPU instruction and buffer images, and executes the resulting plan on either host emulation or RTL simulation. Unsupported work is never hidden; it appears explicitly as host-side steps in the execution plan.

The useful unit of design is the whole stack, not the RTL alone. Compiler-visible tensor semantics, physical packing format, host/NPU boundaries, verification points, and repeated-run runtime behavior all affect whether the accelerator is practical to use. TinyNPU therefore includes explicit host-op dispatch, compiler-owned expected tensors for verification, repeated-run sessions with persistent state, and benchmark accounting that separates accelerator compute cycles from runtime and protocol overhead.

The implemented stack has been validated end-to-end on a trained MNIST model, and a controlled multi-tile matrix multiplication benchmark is used to study compute speedup and integration overhead. In addition, a reproducible Yosys-based synthesis flow provides rough resource estimates for the current design. The report presents TinyNPU as a working accelerator research system for studying architecture, compilation, packing, and workload fit together.

TinyNPU

(ÖZET)

TinyNPU, sistolik dizi tabanlı bir nöral işlemci birimini onu kullanılabilir hale getiren derleyici, çalışma zamanı, niceleme akışı ve benzetim tabanlı doğrulama ortamıyla birlikte ele alan bütünleşik bir donanım-yazılım sistemidir. Amaç yalnızca küçük bir hızlandırıcı çekirdeği tasarlamak değil, bu çekirdeği gerçek bir makine öğrenmesi iş akışında kullanılabilir hale getirmektir. Bu nedenle sistem, PyTorch tabanlı model hazırlama sürecini destekleyen, nicelemeli modelleri hızlandırıcı segmentlerine dönüştüren, desteklenmeyen işlemleri ana işlemci üzerinde çalıştıran ve sonuçları yazılım referanslarıyla doğrulayan bir yapı olarak tasarlanmıştır.

Donanım tarafında TinyNPU, birleşik tampon bellek ve komut belleği ile beslenen sistolik dizi tabanlı bir hesaplama çekirdeği etrafında kurulmuştur. Girdi aktivasyonları ve ağırlıklar birleşik tampondan akıtıcı birimlere, oradan da sistolik diziye iletilir. Hesaplama sonucunda oluşan değerler son işleme biriminden geçirilerek paketlenmiş biçimde tekrar belleğe yazılır. Bu yapı, yoğun nicelemeli matris işlemleri için verimli bir veri yolu sağlarken, dışarıdan görülen programlama modelini de yönetilebilir düzeyde tutmaktadır. Yazılım tarafı ise yüksek seviyeli model işlemlerini hızlandırıcı segmentleri ve gerekirse ana işlemci işlemlerini birlikte içeren bir yürütme planına dönüştürmektedir.

Bu çalışmada sistem, eğitilmiş MNIST modeli dahil uçtan uca iş yükleri üzerinde doğrulanmıştır. Ayrıca, hızlandırıcının gerçek hesaplama çevrimlerini paketleme, arayüz ve çalışma zamanı maliyetlerinden ayıran bir ölçüm altyapısı kurulmuştur. Böylece çekirdeğin kendi verimliliği ile entegrasyon maliyetleri ayrı ayrı değerlendirilebilmektedir. Tekrarlı çıkarım senaryoları için oturum tabanlı çalışma ve ağırlık kalıcılığı desteği de eklenmiştir.

Rapor, bu sistemi donanım mimarisi, model hazırlama ve derleme akışı, doğrulama yöntemi ve deneysel sonuçlarıyla birlikte sunmaktadır. Sonuçlar, sistolik çekirdeğin güçlü hesaplama sağladığını, ancak toplam davranışın hâlâ veri yerleşimi, bölütleme ve çalışma zamanı maliyetlerine duyarlı olduğunu göstermektedir.

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1 Introduction and Project Summary

1.1 Background and Motivation

Modern deep neural networks demand substantial compute and memory bandwidth even at inference time. This has made specialized accelerator design a central topic both for datacenter-scale systems and for smaller edge-oriented deployments [8, 4]. Quantization is one of the main techniques that makes such deployment practical, because it reduces memory traffic and allows arithmetic throughput to increase when narrow integer types are used instead of floating-point data.

However, quantization is not a single fixed target. Mixed-precision work has shown that different layers or kernels inside the same model can tolerate different bit-widths [11]. This creates a hardware opportunity and a software problem at the same time. Hardware should benefit from narrower arithmetic when it is safe to do so, but the software stack must still prepare the model, preserve numerical intent, and decide where accelerator execution ends and host execution begins.

1.2 Problem Definition

The main problem addressed by TinyNPU is therefore not only how to build a small systolic-array accelerator, but how to make mixed-precision quantized computation usable from a realistic model flow. Fixed-precision accelerators leave throughput on the table when different parts of a model have different precision needs. At the other end of the spectrum, CPU-centric approaches based on ISA extensions or soft SIMD can support low-bit computation, but they still pay substantial software-side cost for packing, unpacking, and orchestrating the data movement around those operations [3, 1, 10].

This project treats the problem as a co-design task. The hardware must expose a regular datapath that benefits from packed low-precision arithmetic. The compiler and runtime must make tensor layout, quantization boundaries, and host fallbacks explicit rather than implicit. The result should be a system that can be measured honestly: accelerator compute, runtime overhead, and host-side work should not be conflated.

1.3 Project Scope and Approach

TinyNPU approaches this problem with four coupled components:

1. **A parameterized mixed-precision accelerator:** TinyNPU is centered around an output-stationary $N \times N$ systolic array, a unified buffer, streamer logic that feeds the array, and a post-processing unit that performs bias addition, rescaling, activation, saturation, and packed writeback.

2. **Packed integer computation:** The processing elements support packed INT16, INT8, and INT4 execution modes. Throughput therefore scales with precision reduction, not by changing the global architecture, but by packing more useful work into the same datapath.

3. **A segmented compiler/runtime:** The software flow traces supported PyTorch models, partitions them into accelerator segments and explicit host-side operations, lowers legal accelerator work into instruction and buffer images, and executes the resulting plan on either host emulation or RTL simulation.

4. **Verification and measurement infrastructure:** The project includes compiler-owned expected tensors, runtime verification points, repeated-run sessions, and benchmark accounting that separates accelerator compute from protocol and software overhead.

TinyNPU runs a trained MNIST model through the full compiler/runtime stack and benchmarks a controlled multi-tile matrix multiplication workload. These experiments expose the architecture, packing behavior, compiler boundaries, and runtime cost of the current system.

The rest of this report follows that logic. The next section positions TinyNPU relative to relevant literature in mixed-precision quantization, matrix multiplication, and accelerator design. The system model section then explains the implemented hardware and software stack in detail, followed by the experimental method, the current results, and the most important next steps.

2 Comparative Literature Survey

2.1 Mixed-Precision Quantization

Mixed-precision quantization work is the clearest algorithmic motivation for TinyNPU. Wu et al. formulate layer-wise quantization as a search problem and show that different layers can be assigned different bit-widths without treating the whole model uniformly [11]. This is the key observation behind a variable-precision accelerator: if model layers do not all need the same precision, a fixed-precision datapath cannot exploit the full efficiency opportunity.

Quantization is also relevant as a compiler concern rather than only a hardware concern. SmoothQuant is an example of pushing difficult numerical adjustments into an offline transformation so that runtime execution becomes simpler and more hardware-friendly [12]. TinyNPU follows the same general philosophy even though it does not implement the full SmoothQuant method itself: model preparation, fused rescaling, and compiler-ready conversion are treated as software responsibilities that simplify the accelerator contract.

2.2 High-Performance Matrix Multiplication

The mathematical core of TinyNPU is still matrix-style computation, so the project is strongly informed by classical GEMM literature. Goto and van de Geijn show that high-performance matrix multiplication depends not only on arithmetic throughput, but on careful tiling and data movement through the memory hierarchy [5]. The same principle appears in TinyNPU at a smaller scale. When matrices exceed the dimensions of the systolic array, the compiler and runtime must tile along the M, N, and K dimensions while preserving the data reuse pattern expected by the array and its post-processing path.

2.3 Existing Accelerator Architectures

On the accelerator side, Google's TPU remains the canonical reference for systolic-array-style matrix acceleration [8]. Gemmini is especially relevant because it demonstrates how a systolic array, programming model, and system context can be treated as one full-stack research object rather than as an isolated RTL block [4]. That full-stack perspective is one of the strongest methodological reference points for TinyNPU.

An educational TinyTPU implementation was a useful starting point for understanding systolic-array organization, but it was not a good endpoint for this project [2]. Extending it toward mixed precision, tiled execution, and cleaner post-processing exposed the limits of treating the datapath as the whole problem. TinyNPU therefore diverged into a custom implementation with a different software contract, explicit host/accelerator boundaries, and a compiler/runtime that owns packing and verification rather than

leaving those concerns informal.

Taken together, these references define the design space in which TinyNPU sits. It is not trying to outscale large industrial accelerators. Its niche is a compact mixed-precision accelerator system in which the hardware, packing model, compiler/runtime, and measurement methodology can all be studied together in one working platform.

3 Developed Approach and System Model

The system is described here as four cooperating subsystems: the TinyNPU hardware itself, the quantizer and model-preparation path, the compiler, and the runtime. This organization matches the real engineering boundaries of the stack and makes the movement of tensors, instructions, and packed data explicit.

3.1 System Overview

At a high level, the flow is straightforward. A model starts either as a quantized PyTorch module or as a compiler-ready artifact produced by the quantization path. The compiler traces the model, partitions it into accelerator-supported regions and explicit host operations, lowers legal regions into TinyNPU instruction and buffer images, and packages the result as a compiled artifact. The runtime then executes that artifact step by step, loading the accelerator when needed, invoking host-side operations when needed, and checking expected tensors when verification is enabled. Inside the hardware, the unified buffer, streamers, systolic array, and post-processing unit implement the actual matrix-style execution path.

That end-to-end view is important because the main design problem is not only to build a datapath. It is to maintain a clean contract between logical tensors, physical packed layouts, host-side fallback behavior, and the actual hardware pipeline. The rest of this section therefore explains the system in the same order: hardware first, then model preparation, then compilation, then runtime execution and verification.

3.2 NPU

The TinyNPU hardware is organized around a control path and a compute path. The control path is responsible for instruction sequencing, host service, and control-state progression. In the current runtime path, most UB/IM staging uses the shared-SRAM host window, while MMIO remains the launch/status control mechanism and compatibility fallback. The compute path consists of the unified buffer, streamer logic, a parameterized $N \times N$ output-stationary systolic array, and the post-processing unit. During active computation, data is read from the unified buffer, oriented by the streamers, consumed by the processing elements, and then drained through the post-processing unit before being written back in packed form.

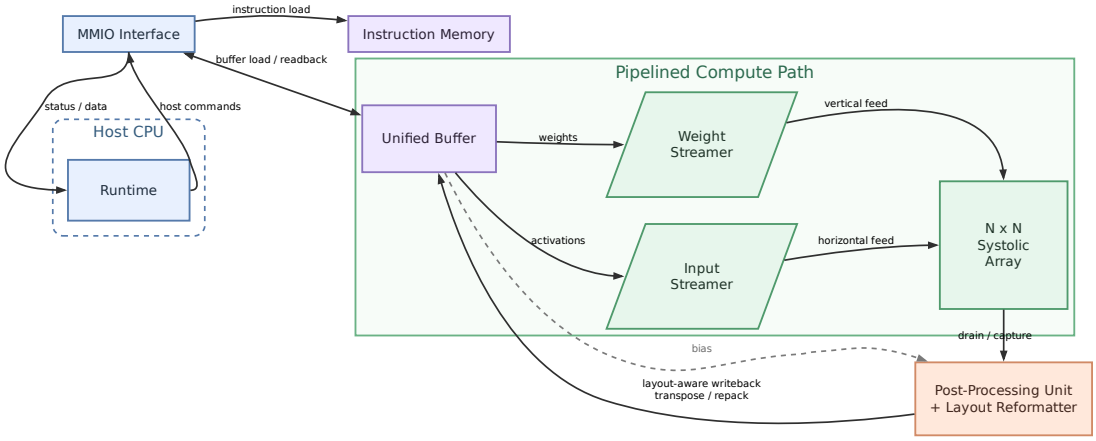

Figure 3.1: Top-level TinyNPU hardware architecture.

Figure 3.1 shows the top-level organization used throughout this report. The host-side runtime communicates with the accelerator through the MMIO interface, which is responsible for instruction loading and buffer access. Inside the accelerator, the unified buffer, input streamer, weight streamer, and systolic array form the main pipelined compute path. The post-processing unit is shown separately because it becomes active during the later drain and writeback phase rather than during the main feed phase of the array. This distinction is important for explaining both data movement and measured execution cost.

3.2.1 Unified Buffer and Physical Roles

The unified buffer is not just a generic scratchpad in this design. Architecturally, it behaves as a true wide dual-port memory with a masked write path. One read side serves the activation stream, while the other serves the weight stream; occasional control-unit random reads are arbitrated onto the activation-side port. This form was chosen to keep both array feed directions active at the same time. A narrower or less concurrent memory interface would turn buffer access into the bottleneck and leave the systolic array underutilized. Because the buffer is wide and row-oriented, this organization is also compatible with a banked memory implementation rather than requiring a single monolithic custom memory block.

Within that buffer, the names A, B, C, and BIAS refer to physical roles in the datapath. A is the left-fed operand: its tiles are packed so that each unified-buffer row can launch one horizontal activation vector into the array boundary. B is the top-fed operand: its tiles are packed for the vertical weight stream. C is the writeback region: it stores quantized output tiles in the packed format chosen by the output precision, so its physical layout may differ from the logical matrix view. BIAS is a separate region consumed only by the PPU. These names therefore describe how data is stored and

consumed by the hardware pipeline, not whether a tensor was logically an activation, a weight, or an intermediate result in the original model. Internal outputs are written back in the physical layout required by their next consumer, including an A-compatible transposed/repacked form for chained matmuls.

3.2.2 Systolic Array and Skewers

The array is output-stationary: packed activation words enter from the left edge and move one PE to the right per cycle, packed weight words enter from the top edge and move one PE downward per cycle, and each PE keeps its output accumulator stationary for the full reduction along the K dimension. Only after the last K contribution has been consumed does the array switch into drain mode and shift those accumulated results downward into the post-processing unit. This choice fits the dominant tiled-GEMM workload well because it keeps the reduction local to the PE and avoids repeatedly moving partial sums through the buffer system during the K loop.

The streamer logic between the unified buffer and the array provides the timing skew that makes the systolic wavefront legal. On the activation side, row i is delayed before it enters the left edge; on the weight side, column j is delayed before it enters the top edge. The values themselves do not change. What changes is when they enter the array. That timing offset creates the diagonal wavefront expected by the schedule, so the activation and weight pieces that belong together meet at the correct PE in the correct cycle. A tile therefore executes in three phases: feed the packed K-steps into the array, allow the wavefront to fill across the remaining diagonals, and then drain the stationary accumulators.

3.2.3 Processing Element

The processing element is where mixed precision becomes concrete. Each PE latches one 16-bit packed activation word and the lower 16 bits of the vertical bus as the current packed weight word. In INT16 mode that word holds one signed multiply; in INT8 mode it holds two signed byte multiplies whose products are summed; in INT4 mode it holds four signed nibble multiplies whose products are summed. The resulting 32-bit partial sum is sign-extended and accumulated into a 64-bit stationary register.

One subtle but important implementation detail is that the vertical PE-to-PE path is 64 bits wide even though the compute datapath consumes only 16-bit packed weights. This is not wasted width. During compute, only `data_from_top[15:0]` is interpreted as the incoming packed weight word and the full 64-bit bus is simply delayed and passed downward. During drain, the same physical column bus is re-used to shift full 64-bit accumulators from one row to the next. The shared bus keeps the PE interface uniform across compute and drain instead of requiring a second dedicated drain network.

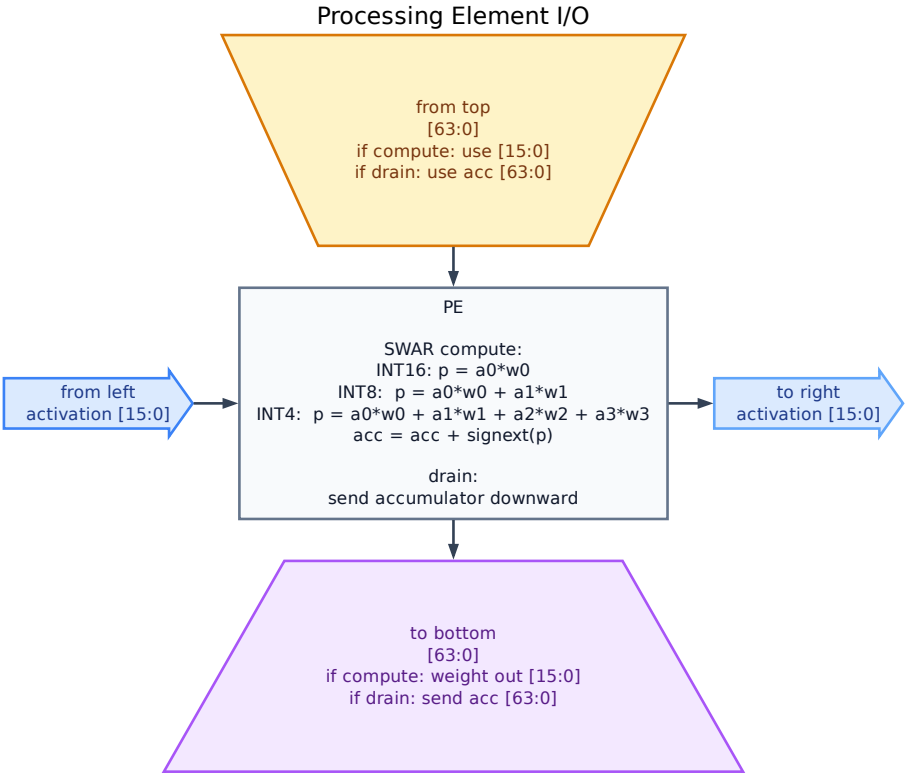

Figure 3.2: Processing-element I/O in compute and drain modes.

3.2.4 Post-Processing Unit

The post-processing unit (PPU) is best understood as an N-lane row processor backed by an internal $N \times N$ tile buffer. During drain, it receives one N-element accumulator row vector per cycle from the bottom of the systolic array, processes all N lanes of that row in parallel, and stores the quantized result into its internal tile buffer. Only after the full tile has been captured does it begin writing buffered rows back to the unified buffer. The current PPU therefore is not a second streaming mesh; it is a row-wise post-processing stage followed by buffered writeback.

The PPU loads one output-column bias vector before drain begins, in two chunks of four 32-bit values. That fetch happens while the tail of the current wavefront is still finishing inside the array, so the bias load is largely hidden under the transition from feed/wait into drain rather than paid as a separate row-by-row cost. For each drained row it then evaluates the same requantization pipeline lane by lane:

$$q = \mathrm{clip}_p\left(\mathrm{act}\left(\left((acc + bias) \times multiplier + 2^{shift-1}\right) \gg shift\right)\right)$$

where acc is the 64-bit drained accumulator, $bias$ is a sign-extended 32-bit bias term, $multiplier$ is the 16-bit fixed-point requantization factor, $\gg$ is an arithmetic right shift, $\mathrm{act}(\cdot)$ selects the configured post-processing nonlinearity, and $\mathrm{clip}_p(\cdot)$ saturates to the

signed range of the selected output precision. In the current stack, the activation stage supports plain ReLU, an integer sigmoid path derived from a DI-Exp-style integer nonlinear primitive in the spirit of I-LLM [7], and a hardware-friendly hard-GELU approximation used when GELU is lowered into the accelerator [6, 9]. Concretely, the datapath performs bias addition, fixed-point multiplication, round-to-nearest when $shift > 0$, the selected activation, and final saturation to INT16, INT8, or INT4.

Writeback is also precision-dependent. After the tile buffer has been filled, the PPU emits one buffered row per writeback cycle. For INT16 output, each 16-bit lane is written back directly. For INT8 and INT4 output, the PPU shifts the quantized value into the byte or nibble slot selected by the packed write offset, and the unified buffer applies a bit mask so that only that subword is updated while previously written sibling tiles in the same physical word are preserved. The control unit derives this packed write offset from the output tile index: two logical M-tiles share one physical word for INT8, and four logical M-tiles share one physical word for INT4. The writeback path also supports layout selection for chained tensors. When the compiler marks an internal output as the next layer's A-side input, the PPU reformats its buffered tile during writeback into an A-compatible packed layout. For INT16 this behaves like a small tile transpose; for INT8 and INT4 it becomes a transpose-plus-subword repack that places the byte or nibble groups where the left-fed streamer expects them. This layout-aware writeback is part of the validated hardware contract.

3.2.5 Control and Instruction Format

This behavior is programmed through a fixed-width 256-bit instruction format in instruction memory. The current `MATMUL` instruction encodes the opcode, base addresses for the A, B, C, and bias regions in the unified buffer, the tile counts along M, K, and N, and the PPU configuration fields for multiplier, shift, activation, input precision, output precision, hard-GELU metadata, and output-layout selection. The control unit therefore does not fetch a stream of tiny scalar operations; it fetches a coarse-grained instruction that describes one tiled matrix-multiplication region and then steps through clear, feed, wait, drain, bias-load, and writeback states internally. This ISA was kept deliberately simple because the supported workload is highly predictable. A finer-grained instruction stream would add fetch and decode overhead without providing much useful flexibility in the current compiler contract.

Table 3.1: Implemented `MATMUL` instruction field layout in the current RTL. The decode path now consumes both hard-GELU metadata and output-layout selection for chained writeback. Remaining low bits are presently unused.

Bits	Field	Meaning
[255 : 252]	`opcode`	Operation code; `ISA_OP_MATMUL` selects the tiled matrix-multiplication sequencer.
[247 : 232]	`A_base`	Unified-buffer base address of A tiles in activation layout.
[231 : 216]	`B_base`	Unified-buffer base address of B tiles in weight layout.
[215 : 200]	`C_base`	Unified-buffer base address of C writeback tiles.
[183 : 168]	`M_tiles`	Number of output-tile positions along the M dimension.
[167 : 152]	`K_tiles`	Number of reduction passes per output tile.
[151 : 136]	`N_tiles`	Number of output-tile positions along the N dimension.
[135 : 120]	`BIAS_base`	Unified-buffer base address of the bias payload consumed by the PPU.
[119 : 112]	`shift`	Arithmetic right-shift amount for requantization.
[111 : 96]	`multiplier`	Fixed-point requantization multiplier applied in the PPU.
[95 : 88]	`activation`	Post-processing activation selector consumed by the PPU; current software uses it for ReLU, integer sigmoid, and hard-GELU modes.
[87 : 86]	`out_precision`	Output packing precision used by the PPU writeback path.
[85 : 84]	`write_offset`	Reserved packed-lane field in the current instruction format. In the present RTL, the effective write offset is derived internally from output-tile index and output precision during packed writeback.
[83 : 82]	`in_precision`	Input precision mode broadcast to the systolic array PEs.
[81 : 74]	`h_gelu_x_scale_shift`	Per-op hard-GELU input-domain shift used by the PPU when GELU is lowered into the accelerator.
[73 : 72]	`output_layout`	Physical writeback layout selector. The current RTL supports ordinary C-style output packing plus an A-compatible transposed/repacked form for chained matmuls.

`HALT` terminates the current instruction stream and places the control unit in the halted state, `NOP` advances without side effects, and `MOVE` is reserved for internal data movement. In the current JIT flow, `MATMUL` is the important emitted operation and `HALT` marks the end of the segment program.

Table 3.2: Implemented instruction opcodes in the current RTL control path.

Opcode	Name	Role
0x0	NOP	No operation; decode advances without performing accelerator work.
0x1	HALT	Terminates execution and leaves the controller in the halted state until the host issues another command.
0x2	MATMUL	Coarse-grained tiled matrix-multiplication instruction with integrated PPU configuration.
0x3	MOVE	Internal data-movement operation reserved by the RTL control path.

Instruction execution is paired with a small MMIO command interface on the host side. The host does not directly drive internal control states; instead it writes a command register to request memory access or to start execution from a given instruction-memory address. In the current runtime configuration used for the refreshed measurements, data-heavy preload/readback traffic is normally served by the shared-SRAM window, and this MMIO command path is primarily used for control and fallback memory service.

Table 3.3: Host-visible MMIO commands used to service the accelerator.

Command	Value	Meaning
CMD_WRITE_MEM	0x01	Writes the current MMIO data value into unified-buffer or instruction-memory space at the selected address.
CMD_READ_MEM	0x02	Reads the selected memory location back into the MMIO-visible data register.
CMD_RUN	0x03	Starts instruction execution from the instruction-memory address carried in the argument register.

3.3 Quantizer and Model Preparation

The quantization path exists to prepare trained models into coherent compiler-ready artifacts. The goal here is not to solve quantized training or bit allocation as standalone research problems. The goal is narrower and more practical: produce models whose weights, activations, ranges, signedness, and optional mixed-precision choices are explicit enough that the compiler can lower them without losing numerical coherence.

The role of QAT is clearer in the activation study of Section 5.4. In the W16A16 example there, the reference MNIST classifier reached 71.54% in FP32, post-training quantization reduced that to 35.53%, and QAT recovered it to 69.98%, which is close to the FP32 reference. In that example, QAT is not merely validating the export path. It is the stage that adapts the trained model to the quantized deployment contract that the final artifact must obey.

This part of the stack also includes the mixed-precision helper. Its purpose is limited and should be stated clearly: it is not a global bit-allocation solver. Instead, it provides

a practical one-layer-at-a-time sensitivity workflow that can generate usable mixed-precision configurations, recalibrate them, and pass the result onward to the compiler-ready conversion path. In other words, it is a way to prepare plausible deployable models for this hardware rather than a claim of optimal bit allocation.

The QAT path is more structured than simply toggling fake quantizers. It has three stages. First, a trained FP32 checkpoint is calibrated to estimate per-layer activation ranges and to initialize activation and weight scales. Second, the model is fine-tuned with quantized layers in place, so activations and weights see fake-quantized values during training. Third, the trained QAT model is frozen into a compiler-ready form with explicit integer weights, scale factors, output precisions, and fused multiplier/shift metadata. In TinyNPU, QAT is therefore the stage that turns a trained floating-point model into a deployable quantized artifact rather than the stage that performs lowering itself.

This is also where accelerator-specific activation approximations are prepared before lowering. For the integer sigmoid and hard-GELU experiments, QAT trains against the same broad activation contract that the compiler later freezes into the deployed artifact. In the GELU case, this includes learning a per-layer hard-GELU input-domain shift that is emitted as instruction metadata for the PPU. The broader direction is motivated by work such as I-LLM and I-BERT, even though the final TinyNPU hard-GELU form is a project-specific hardware approximation rather than a verbatim paper operator [7, 9].

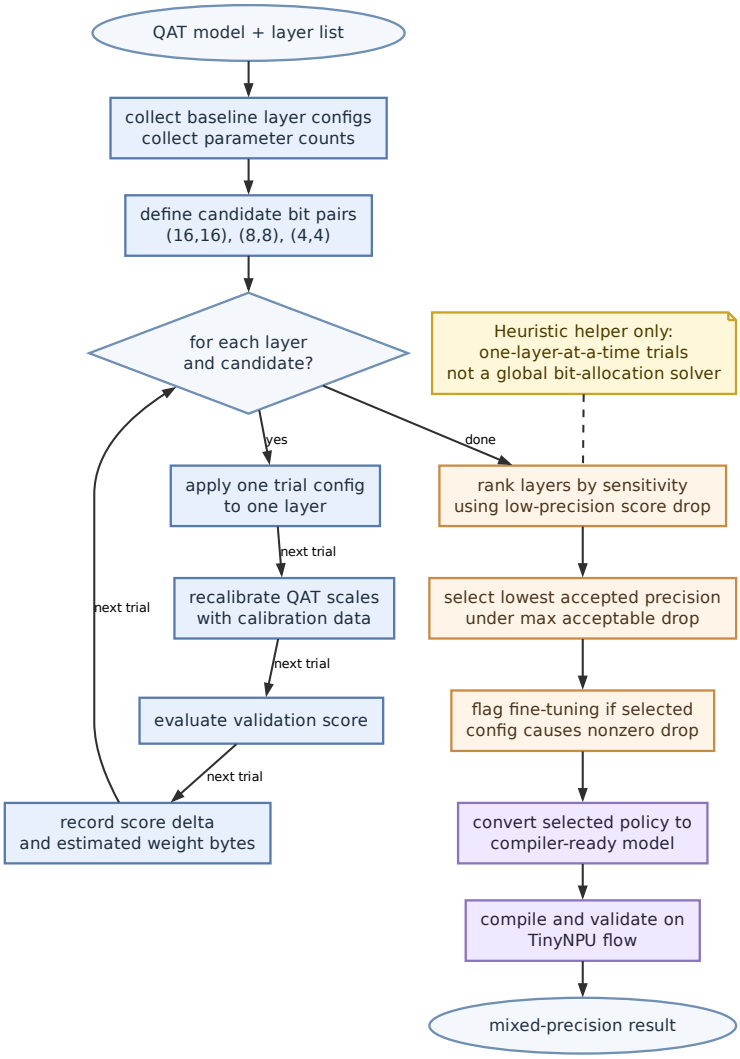

Figure 3.3: Heuristic mixed-precision selection flow used to obtain compiler-ready model variants.

The important role of this subsystem is therefore not to solve quantization research as a whole. It is to provide a disciplined bridge from trained or quantized models into forms that the TinyNPU compiler can lower without losing track of numerical meaning. The resulting artifacts are then carried through segmented compilation and RTL-backed execution, with compiled-host and RTL behavior checked against each other on sampled inputs rather than stopping at an offline export boundary.

3.4 Compiler

The compiler is responsible for converting model structure into a mixed execution structure. Rather than generating a monolithic accelerator-only program for the whole model, the compiler constructs an `ExecutionPlan` whose steps are typed as `NpuSegment`,

`HostOp`, or `VerifyTensor`. `NpuSegment` groups accelerator-supported matrix-style work. `HostOp` keeps unsupported or intentionally host-side work separate. `VerifyTensor` allows compiler-owned expected tensors to be checked later at runtime.

This representation is the main reason the stack remains analyzable. Unsupported behavior is not folded into a fake accelerator-only view, and accelerator boundaries are not implicit. That is a deliberate policy: if someone wants to target TinyNPU effectively, the model and compiler flow should reflect the actual hardware boundaries rather than pretending full-model coverage already exists.

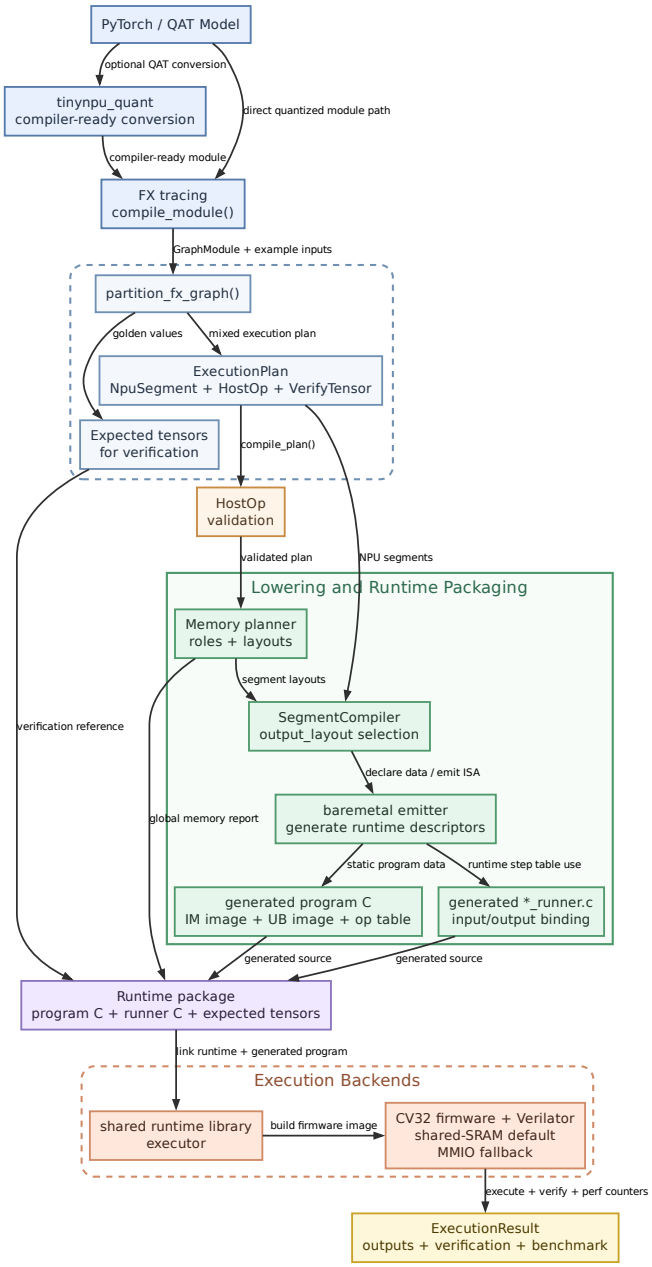

Figure 3.4: Compiler and runtime flow used by the TinyNPU JIT path.

Figure 3.4 summarizes the implemented software flow. A model enters either directly as a supported quantized PyTorch module or through a compiler-ready conversion path built around `tinynpu_quant`. The JIT frontend traces the model with FX, partitions it into a mixed execution plan, validates host operations, and lowers each accelerator segment through the memory planner and `SegmentCompiler`. The resulting `CompiledArtifact` contains the execution plan, per-segment IM/UB binaries, expected tensors for verification, and the memory-layout information required by the runtime. The current emitter/runtime boundary keeps this explicit by generating static program tables in C and executing them through one shared runtime implementation.

Packing is part of the compiler contract as well. The backend uses role-specific physical layouts for activation-side tensors, weight-side tensors, output buffers, and bias payloads. These are physical storage forms rather than semantic tensor meanings, but they are essential because legal execution depends on matching memory layout to the real feed directions and writeback format of the NPU. A useful correction came late in the project: internal NPU outputs can no longer be treated as if ordinary C-style writeback were automatically consumable by a following A-side input. The compiler now annotates the required physical output layout explicitly, and the emitted instruction stream carries that selection into the RTL writeback path.

The important compiler policy is not the cycle-by-cycle tile schedule but where accelerator regions begin and end. The partitioner emits a mixed execution plan and forces a segment boundary whenever execution must return to the host, a verification point is requested, or an operation is outside the current NPU contract. As a result, host-side work such as dequantization, softmax, mean, layout restoration, and verification is represented directly as `HostOp` or `VerifyTensor` steps rather than being buried inside an accelerator launch.

Memory planning makes those execution boundaries explicit in the buffer layout. The unified buffer is divided into a static zone and a dynamic zone: weights and biases receive globally assigned packed addresses that can be shared across segments, while activations and outputs are placed in a per-segment dynamic region with liveness-based reuse. The planner also treats layout as a real physical requirement. If a tensor must be emitted in an A-compatible layout for an internal chained consumer, that requirement is carried in the memory report rather than being inferred informally. At host-visible boundaries, the flow preserves a host-friendly INT16 representation where possible, especially for boundaries that will be read back or consumed by host operations. The reason is simple: when data returns to the host, the runtime prefers to hand off the highest precision the datapath can produce instead of requantizing early and making host-side functions such as softmax or verification unnecessarily lossy.

3.5 Runtime and Verification

The runtime executes the compiled artifact as a mixed host-accelerator loop. For each step in the execution plan, it either launches an accelerator segment, executes an explicit host operation, or checks a verification tensor. The key point is that host-side

work is never hidden. If a step remains on the host, it remains visible in the plan and in the benchmark accounting.

The runtime also owns repeated-run behavior. It can preserve reusable state such as static unified-buffer contents and preloaded instruction-memory images across runs. This does not remove all setup work, but it makes repeated inference more realistic and exposes a measurable optimization path for reducing avoidable overhead.

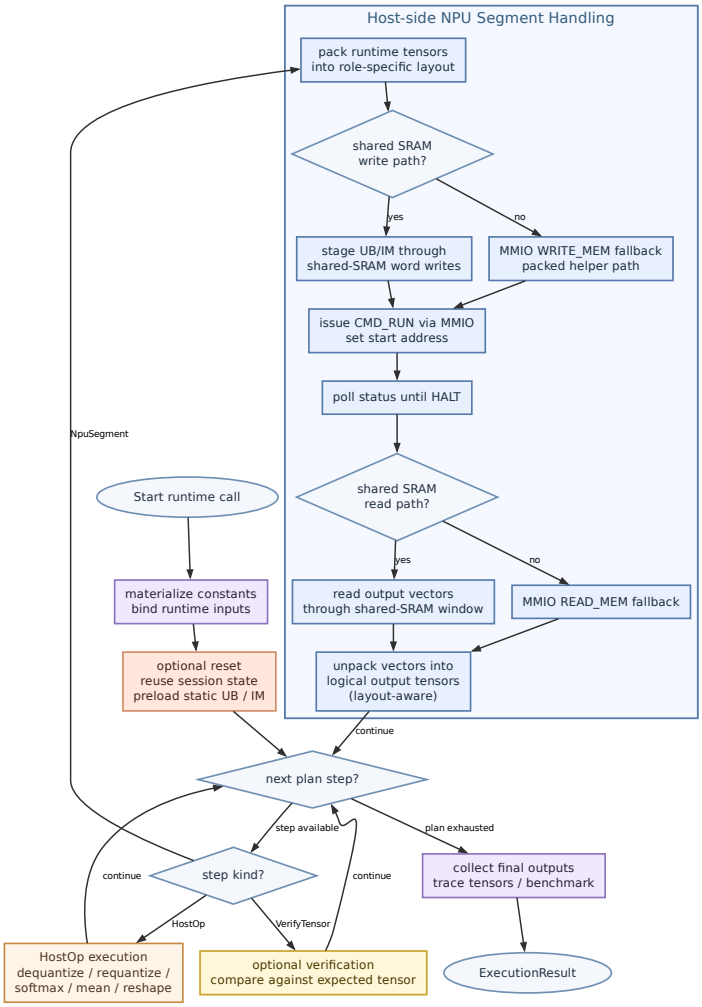

Figure 3.5: Host-side runtime flow for executing a compiled TinyNPU artifact.

Figure 3.5 shows the same loop from the host-side orchestration perspective. For accelerator segments, the runtime first packs logical tensors into the physical layouts expected by the datapath, stages vectors into the unified buffer (shared-SRAM path by default, MMIO fallback available), launches execution, waits for completion, and then reads and unpacks the resulting vectors back into logical tensors. Host-side operations such as requantization, dequantization, softmax, mean, and shape movement appear explicitly in the same loop rather than being hidden behind the accelerator interface. With the newer output-layout support, the host-side runtime no longer has to repair certain internal layout mismatches itself: when the compiler marks a chained tensor as

needing A-format storage, the hardware segment writes it back in that form directly.

Verification is built into this same path. Compiler-owned expected tensors can be checked during execution, which means correctness is not limited to end-of-run output comparison. Benchmarking is also integrated here: the runtime can record hard RTL cycle counts for accelerator run windows while separately accounting for software-side overhead such as packing, MMIO traffic, and readback.

3.6 Current Scope Boundaries

The current system targets a narrow but concrete contract. The mixed-precision helper produces compiler-ready configurations rather than solving the global bit-allocation problem. The compiler supports a bounded operator set and routes remaining work through host steps. The evaluation focuses on one end-to-end model and one controlled kernel benchmark, which is enough to study the interaction between architecture, compilation, and runtime without overstating workload coverage.

4 Experimental Setup and Design

Evaluation uses three evidence sources. Unit and regression tests verify correctness incrementally across the stack, including compiler transformations, execution-plan behavior, memory planning, and host-side operator semantics. RTL-backed cocotb tests then exercise the hardware/software path directly and expose the control-unit performance counters during execution. A reproducible Yosys flow provides rough logic and DSP estimates for the current top-level design, with and without abstracted memories.

Host emulation and RTL simulation execute the same compiled plan, so the same artifact can be checked both for tensor-level correctness and for hardware-visible cost. The control unit reports both total run-window cycles and per-state cycle and entry counts, which makes it possible to separate time spent in feed, wait, drain, writeback, and related control states.

The empirical results are organized around two benchmark views. The first is a controlled multi-tile GEMM chosen to force tiling on the current array and expose the split between array throughput and integration overhead. The second is a model-level case study built from two compiler-ready artifacts that expose how execution-plan shape changes delivered speedup.

The reported metrics keep those cost categories separate. Accelerator cycles are taken from the RTL run window of the NPU segment itself. Packing, MMIO traffic, tensor movement, and host-only operators are accounted for outside that run window. The CPU baselines used later in the report are analytic reference models for the same logical work, not measurements from a physical comparison platform.

For the refreshed runtime-coverage runs in this report, firmware binaries were built with `-march=rv32imfc -mabi=ilp32 -O3` and `-DTNPU_RUNTIME_V2_DUMP_FINAL_OUTPUTS=0`. Runtime validation used `autoverify` against compiler-owned expected tensors when explicit verify ops were not emitted. Simulator runs used `VERILATOR_MAX_TICKS=3000000000` with `+maxcycles=250000` to keep measurements directly comparable.

5 Comparative Evaluation and Discussion

This section focuses on two concrete artifacts: the controlled multi-tile GEMM benchmark in Table 5.1, and the updated fair INT16 conv/linear runs in Tables 5.3 and 5.4. The benchmark metrics are meant to separate core accelerator behavior from host-side cost; the benchmark-cost-model appendix defines the cost model and the reported speedup quantities in detail.

5.1 Kernel-Level Benchmark

Table 5.1: Controlled multi-tile matrix multiplication benchmark results for the 13×17 by 17×19 workload.

Reference model	**CPU replaced cycles**	**NPU run-window cycles**	$S_{\mathbf{pure}}$
Unpipelined scalar	51870	369	140.57×
Ideal issue-1	30628	369	83.00×
Five-stage in-order	43472	369	117.81×

For this 13×17 by 17×19 GEMM, the measured accelerator run window was 369 cycles for the NPU segment itself. Read Table 5.1 together with Appendix B: "CPU replaced cycles" estimates how much scalar work the same GEMM would require, while S_{pure} compares that work against only the measured NPU run window. This is therefore a kernel result, not a full-model result: it isolates raw compute advantage rather than the cost of launching and servicing the accelerator.

The main takeaway from Table 5.1 is that the arithmetic core itself is strong. Even on a deliberately awkward tiled GEMM, the replaced scalar work is much larger than the accelerator run window. The system-level question of how much of that advantage survives packing, MMIO, and host-visible helpers is answered later by the model-level case study, not by this kernel table.

To connect this kernel benchmark to related low-bit GEMM work, Table 5.2 reports the TinyNPU quantities that overlap with the comparison boundary used for Mix-GEMM [10]. The shared boundary is precision scaling in tiled GEMM kernels, not end-to-end software-stack latency. TinyNPU therefore reports RTL-measured GEMM throughput and a region-level speedup under one fixed TinyNPU host model, while Mix-GEMM reports performance and speedups against OpenBLAS on a SiFive U740. The absolute speedup values are not interchangeable, but the precision-scaling trend is.

Table 5.2: Kernel-level TinyNPU GEMM scaling used for the Mix-GEMM comparison boundary. Region-level speedup uses the ideal issue-1 reference model from Appendix B.

Workload	INT16 MAC/cycle	INT8 MAC/cycle	INT4 MAC/cycle	INT8/16 gain	INT4/8 gain	INT4 speedup
64^3	41.24	60.84	79.80	1.48×	1.31×	32.19×
$96 \times 64 \times 96$	41.30	60.98	80.04	1.48×	1.31×	37.18×
128^3	50.22	82.65	122.06	1.65×	1.48×	62.99×

Table 5.2 also makes one important limitation visible. If throughput scaled only with packed lanes, INT8 would deliver 2× the logical MAC rate of INT16, and INT4 would deliver another 2× over INT8. In the present GEMM set, TinyNPU achieves 1.48–1.65× for INT8/INT16 (74–83% of ideal) and 1.31–1.48× for INT4/INT8 (66–74% of ideal). Mix-GEMM reports closer-to-ideal scaling in the same broad sense; for example, their a5-w5 configuration is reported as a 60% improvement over a8-w8 (1.6×), which matches the ideal 8/5 scaling in that step [10]. The gap in TinyNPU scaling is expected for small-to-medium tiles because fill, drain, and control costs remain roughly fixed while the number of logical MACs per compute cycle rises with narrower packing.

The same table can also be read against the array's theoretical peak. For the current 8×8 array, the peak issue rates are 64 INT16 MACs/cycle, 128 INT8 MACs/cycle, and 256 INT4 MACs/cycle. On the largest reported kernel, 128^3, TinyNPU reaches 50.22, 82.65, and 122.06 MACs/cycle, which corresponds to 78.5%, 64.6%, and 47.7% of those respective peaks. This makes the scaling story clearer than gain ratios alone: larger kernels improve utilization by amortizing fill and drain overhead, but utilization still drops as precision narrows because the fixed control and wavefront costs consume a larger fraction of the run while the logical MAC count per cycle rises.

5.2 Updated Fair INT16 Conv/Linear Runs

The previous fair-run block was refreshed from the latest committed INT16 same-shape cycle logs (`runs/PERF_2026-04-08_INT16_INT8_INT4_FAIR.md`). The comparison now uses three directly comparable programs: linear CPU-vs-NPU, convolution CPU-vs-NPU with `conv_stream` disabled, and the same convolution shape with `conv_stream` enabled.

Table 5.3: Phase-level cycle counts for updated fair INT16 runs (same-shape artifacts).

Program	Path	Host Im2Col	Stage	Run	Readback	Seg.NPU	Seg.CPU
cv32e40p	linear_int16_cpu_vs_npu_linear_demo	0	30313	530	3319	38525	265001
cv32e40p	conv_int16_cpu_vs_npu_conv_demo (conv_stream=off)	16003	35616	529	3451	43838	265626
cv32e40p	conv_int16_cpu_vs_npu_conv_same_shape_demo (conv_stream=on)	0	4083	529	3292	12118	–

Table 5.4: End-to-end cycle comparison from the same fair INT16 artifacts.

Program	E2E NPU	E2E CPU	Speedup	Notes
`int16_cpu_vs_npu_linear_demo`	39586	266062	6.72×	$1029 + 32 + 38525$ versus $1029 + 32 + 265001$.
`int16_cpu_vs_npu_conv_demo` (`conv_stream`=off)	60903	282691	4.64×	Includes host `im2col` in both paths.
`int16_convstream_same_shape_demo` (`conv_stream`=on)	34179	–	–	Same conv shape; used to isolate staging/boundary impact.

The key result is that `conv_stream` does not materially change core arithmetic time for this shape (`run`=529 in both conv artifacts), but it removes most boundary cost. The same-shape conv stage term drops from 35616 to 4083 cycles (8.72× reduction), and total segment NPU cycles drop from 43838 to 12118 (3.62× reduction). This is exactly the expected signature of reducing host-visible layout work rather than changing the MAC datapath itself.

5.3 Current Runtime Coverage

The current runtime baseline was re-checked after the packed conv-stream RTL fix (`6ecf1e3`) using one fixed method: `-march=rv32imfc -mabi=ilp32 -O3`, `-DTNPU_RUNTIM` runtime `autoverify` against compiler-owned expected tensors, and simulator launch with `VERILATOR_MAX_TICKS=3000000000` plus `+maxcycles=250000`. The table reports correctness and phase-level cycle counts under that single setup.

Table 5.5: Post-fix runtime coverage refresh for representative INT16/INT8/INT4 artifacts.

Program	Path	Host Im2Col	Stage	Run	Readback	Seg.NPU	Verify
`cv32e40p_int16_convstream_same_shape_v2`	`int16_convstream`	0	3810	530	3387	13707	OK
`cv32e40p_int8_packed_conv_stream_default_v2`	`int8_packed_conv_stream`	0	7309	98	2931	16267	OK
`cv32e40p_int4_conv_stream_same_shape_v2`	`int4_conv_stream`	52988	40004	290	9453	55821	OK
`cv32e40p_int4_packed_conv_stream_default_v2_inrange`	`int4_packed_conv_stream`	0	9176	82	3199	18386	OK
`cv32e40p_int4_packed_conv_stream_default_v2`	`int4_packed_conv_stream`	0	9176	82	3195	18382	FAIL (`@0`)

Three conclusions matter for the current runtime implementation: INT16 direct conv-stream is stable, INT8 direct packed conv-stream is stable, and INT4 direct packed conv-stream is stable when fixtures respect the signed INT4 contract. The one remaining INT4 failure in this matrix is a fixture-validity issue (`xmat` in $[-14, 14]$ instead of $[-8, 7]$), not a datapath correctness issue.

5.4 Activation Study

The activation study used a small MNIST classifier with three hidden convolution layers followed by mean pooling and a final fully connected layer, then replaced the hidden

GELU path with TinyNPU's hardware-friendly hard-GELU. The result was simple. `W8A8` hard-GELU was too lossy: the FP32 reference reached 71.54%, the corresponding `W8A8` PTQ model fell to 12.14%, and QAT recovered only to 54.44%. Keeping the hard-GELU boundary at INT16 instead changed the outcome substantially: PTQ improved to 35.53% and QAT recovered to 69.98%, close to the FP32 reference. Hard-GELU is only viable when the post-activation boundary retains sufficient precision. The full INT16 variant then exposed the next limit by exceeding the current unified-buffer budget, so the practical next step is selective INT16 retention around sensitive GELU boundaries rather than blanket INT16 everywhere.

5.5 Bottlenecks and Interpretation

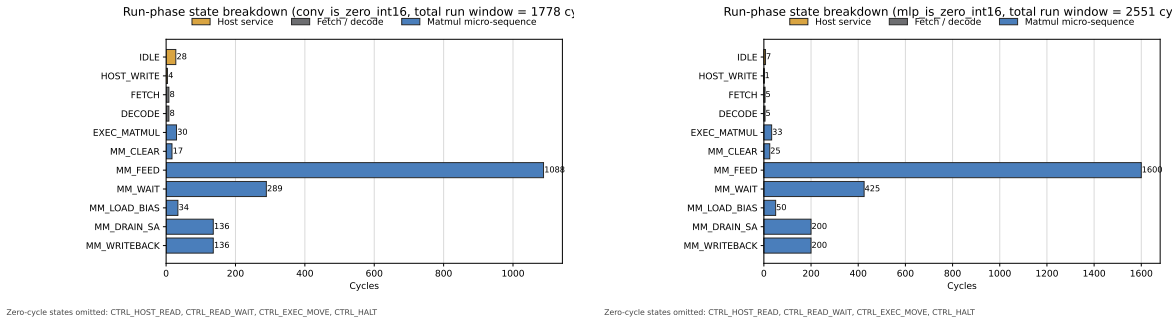

Figure 5.1: Illustrative run-phase control-state distribution for fair conv-like and linear-like artifacts. The figure is used for state-mix interpretation; absolute cycle totals are reported in Tables 5.3 and 5.4.

Taken together, the updated fair-run tables and Figure 5.1 show a consistent bottleneck ranking. Inside the run window, state occupancy is still dominated by `MM_FEED` and `MM_WAIT`, with drain/writeback next and fetch/decode relatively small. The larger practical gains, however, come from boundary-service reduction: in the same-shape conv artifacts, `conv_stream` preserves the run term (529 cycles) but collapses stage and total segment cost. That is why the highest-value optimization direction remains reducing avoidable staging/readback and layout-boundary work, not further increasing raw MAC throughput for these shapes.

A rough synthesis snapshot is included only for scale. Under the Yosys flow used in this report, the top-level design estimates 25464 logic cells when the major memories are abstracted and 33324 logic cells when the current memory implementation is kept, with 227 DSP48 blocks in both cases. In the full-memory run, Yosys also infers 8192 `RAM128X1D` and 944 `RAM256X1S` primitives. These figures are still FPGA-style synthesis proxies rather than final silicon area, but they are sufficient to place the design in a non-trivial hardware range.

6 Conclusion and Future Work

TinyNPU is a working mixed-precision accelerator platform with a usable hardware architecture, a PyTorch-oriented model-preparation flow, a segmented compiler/runtime, explicit host-boundary handling, repeated-inference support, verification infrastructure, and benchmark tooling that separates compute from overhead.

The results show a clear split between compute capability and integration cost. The systolic core delivers strong throughput on tiled GEMM, but practical execution still pays a noticeable cost for segmentation, state movement, packing, and host-visible boundaries. The mixed-precision path is practical for preparing compiler-ready models, and the current evaluation focuses on one end-to-end model and one controlled kernel benchmark.

The immediate priorities are to bring more models onto the stack, reduce avoidable runtime and interface overhead, increase the size of useful accelerator regions, and use measured workloads to justify changes in unified-buffer and instruction-memory sizing. The hard-GELU study sharpened that agenda: richer nonlinearities are feasible in the current PPU, but only if the precision-memory tradeoff is handled explicitly. In the evaluated MNIST GELU study, W8A8 hard-GELU was too lossy, while an INT16-preserving variant recovered nearly all of the FP32 accuracy but exceeded the present unified-buffer budget. That makes selective precision retention around sensitive activation boundaries a more useful next target than simply adding more nonlinear modes. On the hardware side, synthesis and timing characterization should now be extended beyond rough FPGA proxies toward a more complete implementation study. On the software side, compiler work should continue to target the boundary cases and data-movement costs that show up in measured workloads.

A Implemented Activation Algorithms

This appendix records the activation functions as they were actually implemented in the TinyNPU software golden path and mirrored in the RTL primitives. These algorithms therefore describe the deployed integer contract rather than an idealized floating-point reference.

Require: Integer input x_{in}, integer parameters $m_I > 0$, $k_I \geq 0$
Ensure: Integer approximation of an exponential-like decay

$m_F \leftarrow m_I + (m_I \gg 1) - (m_I \gg 4)$
$s_I \leftarrow \left(2^{k_I} + \lfloor m_F/2 \rfloor\right) / m_F$
$t \leftarrow -s_I$
$q \leftarrow \lfloor x_{in}/t \rfloor$
$r \leftarrow x_{in} - q \cdot t$
$u \leftarrow (r \gg 1) - t$
$y \leftarrow u \gg q$
return $\max(0, y)$

Algorithm A.1: TinyNPU DI-Exp

Require: Integer input x_{in}, integer parameters m_I, k_I, output precision parameter p_{out}, optional smoothing factor α_{smooth}
Ensure: Integer sigmoid approximation in a signed p_{out}-bit probability domain

$x_s \leftarrow \lfloor x_{in}/\alpha_{smooth} \rfloor$
$e_0 \leftarrow \text{DI-Exp}(0, m_I, k_I)$
if $x_s \geq 0$ **then**
    $e \leftarrow \text{DI-Exp}(-x_s, m_I, k_I)$
    $n \leftarrow e_0$
else
    $e \leftarrow \text{DI-Exp}(x_s, m_I, k_I)$
    $n \leftarrow e$
end if
$d \leftarrow e_0 + e$
$q_{max} \leftarrow 2^{p_{out}-1} - 1$
return $(n \cdot q_{max} + \lfloor d/2 \rfloor) / d$

Algorithm A.2: TinyNPU DI-Sigmoid

Require: Signed integer input x_{in}, activation-domain shift k_x, slope parameters (n_s, k_s) approximating $1.702 \approx n_s/2^{k_s}$
Ensure: Signed integer hard-GELU result in the same activation domain

$S \leftarrow 2^{k_x}$
$t_3 \leftarrow 3S$
$t_6 \leftarrow 6S$
$s \leftarrow \text{RoundShiftSigned}(x_{in} \cdot n_s, k_s)$
$g \leftarrow \min(\max(s + t_3, 0), t_6)$
$y \leftarrow \text{RoundDivSigned}(x_{in} \cdot g, t_6)$
return y

Algorithm A.3: TinyNPU Hard-GELU

B Benchmark Cost Model

The benchmark path used in this project combines measured RTL cycle counts with an analytic reference-CPU model. The goal is not to claim a definitive comparison against a specific commercial processor. Instead, it is to make the accelerator results interpretable by separating four different questions:

1. How much logical work would a scalar CPU have performed for the same operation?
2. How many cycles did the accelerator actually spend in its measured run window?
3. How much extra cost was introduced by packing, MMIO traffic, and host-side orchestration?
4. How much work still remained on the host because the compiled plan retained helper operations outside accelerator segments?

To support that split, benchmark entries are grouped into four buckets:

1. **CPU-replaced work** ($C_{\mathbf{cpu}}$)**:** logical work that the CPU would have performed instead of an accelerator segment. For matrix multiplication, this includes scalar reads, multiply-accumulate work, loop-control branches, optional bias addition, requantization, and output writes.
2. **Accelerator compute** ($C_{\mathbf{npu}}$)**:** measured RTL cycle counts collected during the accelerator run window with performance monitoring enabled.
3. **Accelerator overhead** ($C_{\mathbf{ovh}}$)**:** software-visible and protocol-visible work required to use the accelerator, such as packing, interface writes, interface reads, instruction loading, and readback.
4. **Remaining host work** ($C_{\mathbf{host}}$)**:** all helper work outside accelerator segments that still executes on the host. This includes `quantize`, `dequantize`, `requantize`, `softmax`, `mean`, `reshape`, `transpose`, `layout_restore`, `im2col`, and alias-style movement helpers.

Primitive work is first counted in an abstract instruction-class form. The counted fields include reads, writes, adds, multiplies, branches, shifts, clamps, bitwise operations, divides, nonlinear helper calls, interface writes, and interface reads.

An analytic CPU model is then applied to those counts:

$$\texttt{cycles} = \sum_i \texttt{count}_i \times \texttt{cost}_i$$

The three reference models currently used in the project are shown in Table B.1. These are implemented directly in the benchmark code and reuse the same counted trace; only the CPI assumptions change.

Table B.1: Reference CPU cost models used in the TinyNPU benchmark flow.

Event class	**Unpipelined scalar**	**Ideal issue-1**	**Five-stage in-order**
Reads / writes	2 / 2	1 / 1	1 / 1
Adds / shifts / clamps / bitops	1	1	1
Integer multiply	4	1	3
Branch	1	1	2
Divide	16	1	16
Nonlinear helper	8	1	8
Interface read / write	1 / 1	1 / 1	1 / 1

Using those symbols, the main reported speedups are:

$$S_{\text{pure}} = \frac{C_{\text{cpu}}}{C_{\text{npu}}}$$

$$S_{\text{full}} = \frac{C_{\text{cpu,full}}}{C_{\text{npu}} + C_{\text{ovh}} + C_{\text{host}}}$$

Here $C_{\text{cpu,full}}$ is the modeled CPU-only cost of the logical program. It starts from C_{cpu} and then adds only host-side functions that would still belong to the logical CPU program, such as `softmax`, `mean`, `reshape`, `transpose`, or an unfused host nonlinearity. Accelerator-serving helpers such as `quantize`, `dequantize`, `requantize`, `im2col`, `layout_restore`, and alias-style movement helpers are excluded from that numerator.

This is why S_{full} is the more useful end-to-end comparison in the current report. S_{pure} isolates the raw compute advantage of the array itself, while S_{full} compares the delivered accelerator execution against a CPU-only execution of the same logical model.

Table B.2: Higher-level helper-op cost models used in the benchmark flow. Here N denotes the number of logical tensor elements and N_{out} the number of reduction outputs.

Helper op	**Count model summary**	$\mathbf{C}_{\text{host}}$	$\mathbf{C}_{\text{cpu,full}}$
`quantize`	reads N, divs N, adds $2N$, clamps N, writes N, branches N	yes	no
`dequantize`	reads N, muls N, adds N or $2N$, writes N, branches N	yes	no
`requantize`	reads N, muls N, adds $2N$, clamps N, writes N, branches N	yes	no
`softmax`	reads $2N$, adds $3N$, divs N, nonlinear N, writes N, branches $2N$	yes	yes
`mean`	reads N, adds N, divs N_{out}, writes N_{out}, branches $N+N_{\text{out}}$ plus optional embedded dequant work	yes	yes
`reshape` / `transpose`	reads N, writes N, adds N, branches N	yes	yes
`layout_restore` / `im2col` / `alias`	reads N, writes N, loop-control/index adds, and branches; `im2col` uses $3N$ adds in the current model	yes	no

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